HealthConnect Analytics Testing & Refinement Report
Week 7: Analytical Testing, KPI Validation & Dashboard Refinement

Track: Data Analytics (cross-track with Data Science)

Dataset: HealthConnect Rwanda database, 27 tables

Date: October 2026

1. Purpose and scope

This report tests the accuracy, consistency and usefulness of the Week 6 analytical outputs. It does not repeat the Week 5 exploratory analysis or rebuild the Week 6 dashboard. It records what was tested, what was found, what changed, and what was re-tested. The objective was to determine whether the dashboard KPIs, calculations, visualizations, and analytical findings accurately represented the underlying HealthConnect dataset. Testing also focused on dashboard filter behavior, unexpected results, data relationship limitations, and the reliability of business insights.

The workbook HC_Clinical_Transactions.xlsx contains six sheets: Appointments, Diagnoses, Laboratory_Tests, Prescriptions, Admissions and Referrals. The Admissions sheet has one row per appointment (5,000 rows) with a length of stay of 1 to 9 days. It has no admitted or not-admitted flag, so admission status cannot yet be defined from our copy of the data. The Referrals sheet covers 4,708 appointments.

2. Testing approach

Testing used four methods, each matched to the type of output being checked.

- Reconciliation: each headline KPI was recalculated independently from the source tables and compared with the dashboard value.Checked record counts, unique identifiers, and relationships between relevant tables.
- Dashboard behaviour: each filter was tested by changing it, checking that visuals and totals respond, and clearing it to confirm the full figure returns.
- Statistical validation: differences and trends were tested with significance tests (ANOVA, Kruskal-Wallis, Kolmogorov-Smirnov, Welch t-test and a linear trend), using Python with the pandas and scipy libraries.
- Timing and leakage review: each model feature was checked for whether it is known at the stated prediction point.
- Reassessed findings relating to appointment attendance and no-shows.
- Shared relevant data validation findings with the Data Science track.
- Documented refinements, retesting outcomes, and remaining issues.

A test passes when the actual result matches the expected result. A failed test is logged with its cause, the fix applied, and the retest outcome. Where a result is not significant, the finding is reworded so the report does not claim a difference the data does not support.

3. Testing register

ID	Area	Test	Expected	Actual	Result	Issue, action and retest
T01	KPI	Total patients	20,000	20,000	Pass	Confirmed unique patients records
T02	KPI	Total billed	9.31bn	9.31bn	Pass	Confirmed unique appointment records
T03	KPI	Billed on matched bills	771,133,439	771,133,439	Pass	NIL
T04	KPI	Total paid	8.39bn	8.39bn	Pass	NIL
T05	KPI	Outstanding balance	919,947,405	919,947,405	Pass	NIL
T06	KPI	Insurance claims	4.88bn	4.88bn	Pass	NIL
T07	KPI	Complaint count	8,518	5,000 shown as 'Negative feedback'	Fail	Replaced with complaints measure (ratings 1–2); retest 8,518.
T08	KPI	Positive feedback	10,583	10,583	Pass	NIL
T09	KPI	Recorded visit days	336	336	Pass	NIL
T10	KPI	Average rating	2.99	2.99	Pass	Average across available feedback records
T11	Display	Emergency visits shown	1,650	2K, then 1.65K	Fail	Card rounding; replaced with text measure; retest 1,650
T12	Filter	Date slicer reaches Emergency visits	Changes with date	Did not change	Fail	Slicer field and link checked; retest: changes with January; returns on clear
T13	Filter	Date slicer reaches Total appointments	Changes with date	Did not change	Fail	As T12; retest pass
T14	Dashboard	Patient experience facility ranking	Varies by facility	2.99 for every facility	Fail	Ranking rebuilt on linked feedback (1,756); retest pass, values 2.76–3.19
T15	Dashboard	Department ranking	Varies by department	2.99 for every department	Fail	As T14; retest pass
T16	Data	Revenue summary table reconciles to billing	≈9.31bn	1.13bn; 60,809 appointments	Fail	Removed from dashboard and from all findings
T17	Analysis	Facility rating differences	Significant if real	p = 0.79 (ANOVA), 0.80 (Kruskal-Wallis)	Not significant	Ranking claims removed; system-wide framing adopted
T18	Analysis	Department rating differences	Significant if real	p = 0.29 (ANOVA)	Not significant	As T17
T19	Analysis	Matched vs unmatched bill amounts	Similar	KS p = 0.66; status and severity shares within one point	Pass	Billing findings generalised with caveat
T20	Analysis	Monthly billing trend (matched)	No sustained trend	Slope -0.3M/month, p = 0.37; variation 5.9%	Pass	Stated as no sustained trend
T21	Analysis	Linked vs unlinked feedback	Similar	Mean 2.93 vs 2.99 (p = 0.05); complaints 40.9% vs 40.4%	Pass with caveat	Caveat recorded

ID	Area	Test	Expected	Actual	Result	Issue, action and retest
T22	Data	Admission records in source	Identified	Admissions sheet found: 5,000 rows, length of stay 1–9; no admitted flag	Correction	Earlier 'no admission records' statement was wrong; admitted definition requested
T23	Attendance	Attendance or no-show field	Identified	None in dataset	Cannot test	Recorded as limitation (section 8)
T24	Filter	Status slicer	Filters cards and charts; totals reconcile	Cards and charts respond	Pass	Retested
T25	Filter	City slicer	As T24	Cards and charts respond	Pass	Retested
T26	Filter	Location slicer	As T24	Cards and charts respond	Pass	Retested
T27	Filter	Gender and Province slicers	As T24	Cards and charts respond	Pass	Retested
T28	Filter	Facility and Department slicers	As T24	Cards and charts respond	Pass	Retested
T29	Filter	Rating slicer	As T24	Cards and charts respond	Pass	Retested
T30	Visual	Map visual displays provinces	Bubbles at province centres	Displays when online	Pass	Retested
T31	Visual	Currency consistent across pages	One currency throughout, from the countries table	Single currency adopted	Pass	Label standard (RF or RWF)

4. KPI validation

All headline KPIs were recalculated from the source tables. Totals, billed and paid amounts, outstanding balance, claims, complaint counts, average rating and recorded days all match the reference values (T01 to T10). Two display-level failures were found and corrected (T07 and T11). The KPI log is kept separately and is referenced here.

5. Analytical validation results

5.1 Billing representativeness

The 5,000 matched bills were compared with the 55,000 unmatched bills (T19). Their amount distributions are similar (Kolmogorov-Smirnov p = 0.66), and payment status and diagnosis severity shares differ by under one percentage point. This indicates that billing and payment records are linked consistently through BillingID, but the relationship between billing and appointments requires further investigation. The difference between total billing and total payments was approximately RF919.95 million. Billing findings can therefore be generalised beyond the matched subset, with that caveat stated.

5.2 Monthly billing trend

A linear trend across the twelve monthly totals for matched bills gives a slope of -0.3 million per month (p = 0.37). Monthly variation is 5.9% of the mean (T20). The data shows no sustained growth or decline.

5.3 Patient experience

The patient feedback dataset contained 21,064 records, with an average rating of approximately 2.99, and a complaint rate of 40.9%, . Only 1,756 feedback records matched appointment IDs in the clinical transactions dataset. This limits the representatives of appointment-linked departmental feedback analysis. The dashboard's overall feedback average should therefore be interpreted as an aggregate across available feedback records, not as a date-filtered measure of appointment satisfaction.

5.4 Insurance records

The insurance claims dataset contained 42,036 records, exceeding the 20,000 unique patient records. Consequently, the number of claims or insurance-related records should not be presented as the number of unique insured patients. An insurance identifier or claim record also does not independently establish active insurance coverage or successful claim processing.

6. Dashboard testing evidence

6.1 Date slicer testing

The dashboard date slicer was tested using two date ranges:

- January 1 to December 28, 2026.
- January 1 to June 30, 2026.

Appointment-related charts, including monthly clinical visits and department visit counts, responded to the selected date range.

However, some KPI cards did not respond as expected.

Total Appointments KPI

The original Total Appointments card did not respond correctly to the date slicer, even though the underlying measure was correct. The original card was replaced with a fresh card, after which the KPI responded as expected.

Total Patients KPI

A new measure was created to show distinction between

Total Patients: all unique patients in the patient dataset.

Patients Seen: distinct patients with appointments within the selected date range.

Total Billed Revenue filter behavior

Initially, the appointment date slicer did not filter the billing KPI because the relationship's single-direction filter setting did not propagate the appointment filter to billing. The relationship direction was changed to Both, after which the KPI responded to the date slicer. The resulting appointment-linked billing amount was approximately RF771.13 million for the January–December range and RF391.78 million for January–June. Filter responsiveness improved, but the relationship and resulting measure require further validation before the filtered amount can be presented as total clinic revenue.

7. Refined dashboard and visualisations

Changes made as a result of testing, with the before and after state recorded:

Change	Before	After	Reason
Date slicer field	Did not filter clinical cards	Filters Emergency visits and Total appointments	T12, T13
Number display	2K and 1.65K shown	1,650 and 5,000 shown	Card rounding; T11
Negative feedback card	5,000 (not traceable)	Complaints, 8,518	T07
Positive feedback card	Basis unlabelled	Labelled Good and Excellent, 10,583	T08
Patient experience rankings	2.99 for every facility and department	Linked averages, 2.76 to 3.19 for facilities	T14, T15
Ranking interpretation	Presented as a ranking	Note that differences are not significant	T17, T18
Revenue summary chart	Showed 1.13bn trend	Removed; matched-bill trend used	T16
Currency	RF on some cards, none on others	[pending decision]	T31

8. Validated key findings

- Billing: 9.31bn billed across 60,000 bills; 771,133,439 on the 5,000 matched bills, which are representative of the full base.
- Outstanding balance: 919,947,405 across 11,887 bills, about 10% of billed.
- Monthly billing: no sustained growth or decline on the matched subset.
- Payments: 8.39bn, split evenly across card, mobile money, insurance and cash.
- Complaints: 8,518 of 21,064 responses (40.4%) are rated 1 or 2. The rate is similar for linked and unlinked feedback.
- Ratings: no statistically significant difference across facilities or departments. The problem is system-wide, not site-specific.
- Clinical volume: 5,000 appointments with clinical detail; visit types and severity are evenly split; one in three visits is an emergency.
- Staffing: 1,250 staff, 2,000 beds recorded as available. Occupancy and usage are not recorded.

9. Updated business insights

There are insights change from Week 6 as a result of testing.

- Patient experience is a system-wide problem: the earlier suggestion that the lowest-rated facilities or ICU needed targeted action is withdrawn.
- The billing findings hold for the full base, because the matched subset is representative, so the report can state them with a caveat rather than as subset-only findings.

- Admission can now be explored, because the Admissions sheet exists, but the admitted definition is not yet confirmed.
- Departmental feedback averages varied, but only a small proportion of feedback records could be linked to appointment records.
- Department and facility rating differences were not statistically significant in the earlier analysis. These findings should not be used to claim that facility or department performance is definitively equivalent.
- Clinical activity counts, diagnosis records, and feedback records represent different units of analysis and should not be compared as though they measure the same population.

10. Attendance and no-show reassessment

The validated Appointments sheet contained AppointmentID, PatientID, DoctorID, FacilityID, DepartmentID, Date, and VisitType. It did not contain an attendance status, appointment completion status, or no-show indicator. Consequently, attendance and no-show rates could not be independently calculated or validated from this sheet. Any previous attendance-related conclusions should therefore be treated as unverified unless supported by another approved source containing actual attendance outcomes. A reliable attendance analysis will require an appropriate attendance status field or another validated source of appointment outcomes.

11. Cross-track testing contribution (Data Science)

The Data Analytics track reviewed the admission-model features against validated findings, set timing and leakage rules, and shared validated findings with the Data Science track. Records are kept as follows.

11.1 Data Analytics and Data Science Intern 1

Component reviewed: HighAcuityEmergency classification model.

The Data Science intern used a proxy target defined as an emergency visit combined with a severe diagnosis. The initial model showed a hold-out AUC of 0.479, with a training AUC of 0.75 and validation AUC of 0.51. After feature reduction and regularized logistic regression, the reported hold-out AUC was 0.455, while the training and validation AUCs were 0.55 and 0.49 respectively. The refinement reduced the training-validation performance gap, but predictive performance remained around chance.

Data Analytics contribution: Reviewed the feature set, highlighted data availability and interpretation limitations, and reinforced that the target represented a high-acuity emergency proxy rather than confirmed admission.

Outcome: The model was not established as suitable for deployment.

11.2 Data Analytics and Data Science Intern 2

Component reviewed: Hospital admission classification model.

The intern clarified that the target was derived from the Admissions sheet linked through AppointmentID, with 586 admitted cases and 4,414 non-admitted cases in the integrated dataset.

Data Analytics reviewed the proposed features and recommended:

- Calculating Age as of the appointment date.
- Holding Diagnosis and Severity pending prediction-time clarification.
- Testing PriorVisits using only historical appointments.
- Testing appointment month and weekday.
- Retaining FacilityID and DepartmentID as candidates with generalization caveats.
- Further validating MaritalStatus, Education, BloodGroup, and BMI.

The intern subsequently reported excluding Diagnosis and Severity, recalculating Age, and testing time-safe historical and calendar features.

Reported retest results:

Hold-out ROC-AUC: 0.5189.

Five-fold cross-validation ROC-AUC: 0.5237 ± 0.0246.

The intern did not recommend deployment because performance remained close to chance.

Data Analytics contribution: Provided feature validation feedback and identified potential leakage and prediction-time availability concerns.

Outcome: A modest numerical improvement was reported, but the model's predictive performance remains insufficient for deployment. Final target, feature list, hold-out usage, and hold-out class distribution confirmations remain outstanding.

12. Limitations

- Only 5,000 of 60,000 appointments link to facility, department, clinical detail and admissions.
- Bills have no dates, so the monthly trend uses the matched subset.
- There is no attendance or no-show status.
- The admitted flag is not defined in our copy of the Admissions sheet. Admission-based findings are blocked until it is confirmed.
- Facilities are identified by ID only; facility names are not available.
- The high-acuity proxy is defined only on the 5,000 appointments with emergency visit type, giving 544 positive cases. Its confidence interval is about ±0.04 on AUC, so small real effects cannot be detected.
- The difference between billing and payment totals has not been fully reconciled.
- Some segment-level conclusions require additional testing.

- Data Science retest details remain partly unconfirmed.

13. Updated recommendations

Each Week 6 recommendation was reassessed against the testing results.

1. Reconcile billing and appointment records to establish the correct relationship and financial reporting grain.
2. Validate outstanding balances at the BillingID level before using the billing-payment difference as a confirmed outstanding balance.
3. Improve feedback-to-appointment linkage and distinguish overall feedback from appointment-linked satisfaction.
4. Standardize KPI definitions and document the source tables, calculation logic, and filter behavior.
5. Obtain reliable attendance and no-show status data before reporting attendance performance.
6. Review clinical and operational measures according to their actual record grain.
7. Retain facility and department identifiers as analytical candidates while testing their generalizability.
8. Continue model evaluation only after target definitions, prediction-time availability, and evaluation procedures are confirmed.

14. Week 8 recommendations

- Confirm the admitted definition with the data owner, then re-test any admission-based analysis.
- Add attendance status to future data collection, so attendance findings can be tested in Week 8.
- Use the validated findings in the final integration: billing trend, outstanding balance, complaint rate, and the system-wide framing of patient experience.
- Record all cross-track replies and their dates in the evidence file, including any request still outstanding.
- Improving appointment attendance data capture.
- Confirming the reliability and intended use of Data Science model outputs.

15. Conclusion

Week 7 testing identified important strengths and limitations in the HealthConnect analytical outputs.

The validation confirmed several key dataset counts and aggregate measures, while dashboard testing identified filter behavior issues and differences in how registered patients, patients seen, total billing, and appointment-linked billing should be interpreted.

The review also highlighted limitations in attendance data, financial reconciliation, and feedback linkage.

Cross-track collaboration with Data Science contributed to feature review, leakage control, and model refinement. However, the reported model results remain insufficient to support deployment.

Overall, the Week 7 work improved the transparency and interpretation of HealthConnect's analytical outputs and identified important data-quality and integration priorities for Week 8.